BullSO 2026 Invitational

Codebusters C

Bull Science Olympiad at the University of South Florida

Saturday, January 24th, 2026

Team Name:_____

Participant Names:_____

Total Score:_____/xxx **Rank:**_____

Science Olympiad — BullSO Div C 2026

Names of participants: (Please print neatly)

_____

_____

_____

School Name: _____ Team Number: **C**

Check your team: ☐ Varsity ☐ JV1 ☐ JV2 ☐ JV3

Warning: Do not open this packet until given permission to do so.

Note: There are useful notes after this page.

Scoring: Time to solve first problem: _____ (use to calculate Bonus below)

Question	Value	Incorrect letters	Deduction	Score
Timed	292			
1	333			
2	226			
3	542			
4★	197			★
5	239			
6	186			
7	369		≡	
8	192		≡	
9	201			
10★	88			★
11	578			
12	226			
13	292			
14	332			
15	313			
16	246			
17	495			
18	253		≡	
19	254			
20★	345			★
21	478		≡	
22	278			
Bonuses	**Timed:**	**Special Bonus: ★ = 150; ★★ = 400; ★★★ = 750**		
Final Score				

Timed Question **[292 points]** Hidden in the margin of a library book on black holes: decode this Aristocrat by Henry Thoreau. What does it say? When you have solved it, raise your hand so that the time can be recorded and the solution checked.

AOZURZUE NJGCO NZGS J VZUGL VYBGS ZI GSC MYWA

HCOIZYU YQ ITZML. DYGS VJRC VC BUMYVQYOGJDWC ZU J

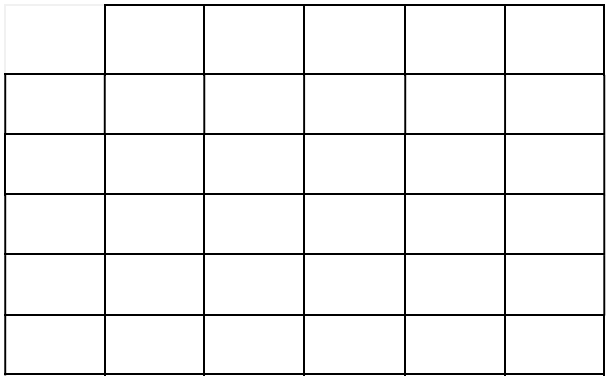
NJL Z OCITCMG DBG MJU'G CUKYL.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	2	3	9	3	1		10	1	4	7	1	4	5	3	5		2	2	4	2	8	5	2		8	9
Replacement																										

1) **[333 points]** Someone left a paper with the message below on the copy machine in the office. It's a quote by Martin Luther King Jr, encoded as a Checkerboard with a polybius key of **POSTPULMONARY.** What does it say?

AO NC LO AO AE LC ON NN NO OO

NO LC ON LR ON LE ON LC AO LO



LN OR AC OC NC AE AE LR AO NC LO AO OC AE AN LR OO NO LC GE

ON LN AO NC NO AC ON OR ON LR NO OO AC AC OC LO LN ON OO LE

ON

2) **[226 points]** This string of symbols appears on the display of a malfunctioning calculator. It hides a quote by George Washington, encoded as a Complete Columnar. The quote has **ACADEMIC** somewhere in it. What does it say?

WCTOT UXOUI IBAXL ASLTS XCHES ETEHO MSYHR STDEK

RMOTC SENXI ARWEM X

Answer: _____

3) **[542 points]** A phrase by Dalai Lama has been encoded using the Patristocrat Cipher using a K2 alphabet. You are told that the cipher contains **civilization**. What does it say?

VLJGX BGERK XBNLH EZFVL JGXBG XGHFJ HEIJS SPRHL

YIXKZ TZHFR PRSXB HEZZB HXFZE XGHLF PLIEJ ARBUX

BWRBW ALWZF BVXKX SXQRH XLB

Replacement																									
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	2	10			6	6	6	9	3	5	3	8		1		3	1	7	4	1	1	3	3	13	1	7

★(Special Bonus Question) 4) **[197 points]** Your smart fridge started flashing this cryptic message at 3 a.m. It's by Bill Gates encoded in a Baconian. What does it say?

Ḷ̈ëą̈ṛ̈n̈L̈ëą̈ṛ̈n̈L̈ëäṛ̈n̈Ḷ̈ëäṛ̈n̈L̈ë̤äṛ̈n̈L̈ë̤ą̈ṛ̈n̈L̈ëäṛ̈n̈L̈ëą̈ṛ̤n̈L̈ë̤ą̈ṛ̤n̈L̈ë̤ą̈ṛ̤n̈L̈ëäṛ̤n̈

L̈ë̤äṛ̈n̈L̈ëą̈ṛ̈n̈Ḷ̈ëą̈ṛ̈n̈L̈ë̤äṛ̈n̈Ḷ̈ëäṛ̈n̈L̈ëą̈ṛ̤n̈L̈ë̤ą̈ṛ̈n̈L̈ë̤ą̈ṛ̤n̈Ḷ̈ëäṛ̤n̈L̈ëą̈ṛ̤n̈L̈ë̤äṛ̈n̈Ḷ̈ëäṛ̈n̈

Ḷ̈ëäṛ̤n̈L̈ë̤äṛ̈n̈Ḷ̈ëäṛ̤n̈Ḷ̈ëą̈ṛ̈n̈L̈ëą̈ṛ̈n̈L̈ë̤äṛ̈n̈Ḷ̈ëäṛ̈n̈L̈ëäṛ̤n̈

5) **[239 points]** Archaeologists unearth a stone tablet inscribed with a message. It's a quote by Albert Einstein encoded as an Aristocrat using a K1 alphabet. What does it say?

1111 GSFVWHFWQR ZO 1111 ZQWYB 1234321 WK VCQ

GUVCQGUVWTUF QISWLUFQYV XA MCWVQ FWBCV RWKHQJKWYB

VCJXSBC U HJWKG VX GUEQ U JUWYZXM.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	4	6		1	5	5	3	1	4	4	1	2		1		9	2	3	1	8	9	12	4	4	3
Replacement																										

6) **[186 points]** You found the following carved into the bark of a hollow log. You are told that it's a quote by Mahatma Gandhi encoded as a Hill with a keyword of ENVY. What does it say?

$$\begin{pmatrix} E & N \\ V & Y \end{pmatrix} \equiv \begin{pmatrix} 4 & 13 \\ 21 & 24 \end{pmatrix}$$

W	E	H	R	T	M	U	O	Q	L	E	V	F	R	I	C	I	Z	A	G	Q	W

7) **[369 points]** An old robot prototype suddenly prints this encoded text. It's a quote by Giannis Antetokounmpo, encrypted using an Aristocrat using a K2 alphabet. What was the K2 Keyword used to encode it?

Enter the Keyword here

Cipher:

LAMVNVLT GAYH KHQJMUVLT VE J MQHHVKZQ VLNALCQLVQLNQ.

QESQNVJZZG OYHVLT JL VBSAHMJLM MQEM ZVXQ MUVE.

Replacement																									
K2	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	1	1		4		2	6		5	2	10	8	4	1		10		2	3	2	12		1	2	4

8) **[192 points]** Found inside a dissected frog's stomach (plastic, don't worry): this Cryptarithm note. What do the values **6798 604** decode to?

Values to decode for solution

6 7 9 8 6 0 4

Cryptarithm formula

```
  L A R R Y
+   B I R D
-----------
J O R D A N
```

	0	1	2	3	4	5	6	7	8	9
A										
B										
D										
I										
J										
L										
N										
O										
R										
Y										

9) **[201 points]** B Lou offers an observation that has been encoded as an Aristocrat using a K1 alphabet. What does it say?

ROT HR PBIT PTTQM QBDT PBIT ONEWTUM, WNH MBJ HR HTO

PTTQM QBDT PRNU ONEWTUM. B YRO'H DORS SAK, WNH BH'M

YTPBOBHTQK HUNT.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	9		3	2			9	2	1	2		6	6	7	6	5	6	2	15	4		4		2	
Replacement																										

★(Special Bonus Question) 10) **[88 points]** An explorer's journal mentions a “hidden truth” encoded below. It turns out to be a quote by Winston Churchill, written in a Porta Cipher using a keyword of **AGONY**. What does it say?.

E	L	H	A	N		R	U	K	O	G		G	X	Y	S	C		F	D	H	C	Q		E	B	U	A	Q		Q	C	H	W	Z

B	V	U	R	X		G	Y	T	X	U		G	C	M	L	I		R

11) **[578 points]** Someone left this encrypted quote scribbled on the chalkboard before class began. It's a quote in Spanish, written as a Xenocrypt using a K1 alphabet. What does it say?

UZ TIESZ VXMRKIZ LTX IXSXREWZR XI WT NEYZ, XR

ZLTXUUZ LTX YXHTXRWMZ LTX WX IXSXREWZ XI UZ RTPZ.

K1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	Ñ	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency					4			1	6		1	4	2	1			1		7	3	8	4	1	5	15	2	11
Replacement																											

12) **[226 points]** Found taped under a desk in study hall: this Nihilist Substitution puzzle by Marie Curie with a keyword of **LUNCHES** and polybius key of **AUTHORISED**. What does it say?

85 24 64 76 55 48 57 82 24 67

55 27 46 55 76 35 54 53 38 55

47 54 25 67 53 27 38 34 84 64 57 54 27 48 38 84 36 66

	1	2	3	4	5
1					
2					
3					
4					
5					

13) **[292 points]** A message flashes briefly on a malfunctioning computer screen. It turns out to be a quote by Ada Lovelace written in an Aristocrat. What does it say?

QBNEAPX N XTTV GTPX TP KQB ENVAT AG UALB GBBAPX N

JAUV NPAONU AP AKG PNKYENU QNIAKNK, IYK QBNEAPX AK

AP N ZUNCUAGK AG UALB GBBAPX AK AP N RTT.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	19	9	1		4		7		2	1	9	2		14	1	11	4	1		7	7	3		6	2	1
Replacement																										

14) **[332 points]** You receive an anonymous text message containing a coded passage. It's said to be a quote by Curt Cignetti, encrypted with a Complete Columnar. The quote has **OPPOSITE** somewhere in it. What does it say?

NASET LEMAM OHUBH EREEE DAASR EOOEA RLEEP NLNEE

DNPTB DTNGB THEEE EDYIW RURI

Answer: _____

15) **[313 points]** This code is found written in chalk on the sidewalk outside the science building. It's a quote by Charles Darwin, written in a Baconian Cipher. You are told that the deciphered text ends with **nest.** What does it say?

AN ARK AGAIN CARRY AN EGG CAMEL AN ELK ACTED ALONE

CAUSE ARDOR ANGIE CAUSE ANGRY ALLAN ASKED CENTS VISOR

BACKS VERGE SAMMY UNDER RANCH IS ANY LEMON JULIO

RODEO HAVEN MASKS VOCAL PETEY MASKS FUZZY HAVIN FINDS

RHYME LEROY ELIZA GRAIL LEFT A RODEO FRAUD SEALS

MOURN ROUTE FEEDS FEVER JOHAN METER KNEEL LABOR

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Replacement																										

16) **[246 points]** A coded message appears after you shine a UV light on a classroom poster. It turns out to be a quote by Henry Ford, encrypted using a Nihilist Substitution with a keyword of **MANUEL** and polybius key of **PROMULGATED.** What does it say?

38 58 68 28 68 42 68 55 66 46

36 44 26 47 66 47 38 36 38 57

66 36 46 62 57 45 84 60 47 46 38 47 84 58 47 56 29 35 67 28

49 56 39 35 95 56 70 46 55 47 88 27 50 44 35 44 97 49 40 64

	1	2	3	4	5
1					
2					
3					
4					
5					

17) **[495 points]** You receive this cryptic message from the future through an online time capsule. It's a quote by Sun Tzu, written in a Baconian. What does it say?

rdoayudagyhdaroughydrougharydouaghryougdhraydoayud

gahyrougdahyrougdahrouydgahroydugayhdaroughydrough

arydoughraoughrougyhrdouaghyrdaoughrydoauygdahryda

ouyghdrougahryodughraouydgharouydaghydrayoudghraou

gydhraouyghdroughrougahyr

18) **[253 points]** Solve this quote that has been encoded using the Cryptarithm Cipher. What do the values **34516 30862** decode to?

Values to decode for solution

3 4 5 1 6 3 0 8 6 2

Cryptarithm formula

```
  A R C T I C
    F R O S T
-
-------------
  C H I L L
```

	0	1	2	3	4	5	6	7	8	9
A										
C										
F										
H										
I										
L										
O										
R										
S										
T										

19) **[254 points]** Srinivasa Ramanujan offers us some advice that has been encoded as an Aristocrat. What does it say?

OUEOVKEQQT, YORFQGCB XFCAT KUIZ F LFLT GY ARKGCGORQT

CIO FY RFYT FY GO YIECAY. OUEYO ZR. G YHRFJ KUIZ

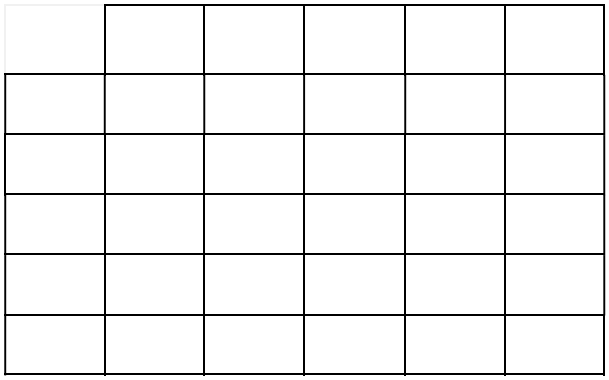
RDHRUGRCXR.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	3	1	6	1	4	8	7	2	4	1	4	2			8		4	10		5	5	1		2	9	3
Replacement																										

★(Special Bonus Question) 20) **[345 points]** Gordon Ramsay has been heard to say the following phrase that has been encoded using the Checkerboard Cipher with a polybius key of **SUBCONTIGUOUS.** What does it say?

NA ME NA MD ME NE ES MS NH AH

NS NS MS MS EA ED ES EA AA EA



MD EE ES EH NS SD MA AH EH EH AH NE EH AD EE ES AD ME SH EA

EH EH EE MS EH AH MS AH MS NA AH MD EA EE NS NS SD EE EH EH

AD AH NH ME SS EA AH MS

21) **[478 points]** Someone passed you a piece of paper with this Aristocrat encoded quote by Mother Teresa using a K3 alphabet. What was the K3 Keyword used to encode it?

Enter the Keyword here

Cipher:

FQG BZEJGY GSPQF ATGHB'F YGKGSDG GBTZPQ KYGASF LTY

KYGCFSBP GDGYI TFQGY BZEJGY TY ASPSF TB C HKTYGJTCYA.

K3	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	4	6	3	2	2	7	15	2	1	3	4	1				4	4		6	8					11	3
Replacement																										

22) **[278 points]** While scrolling through an old floppy disk labeled "TOP SECRET", you discovered this quote by Terry Davis hidden as an Aristocrat. What does it say?

TMZIQ RZHAME NXVB ZI QKY HNXVV ZV JIM JS BWM YJVB

MYTXPPXVVZIQ BWZIQV X NZFZIQ JPQXIZVY HXI MIELPM.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
Frequency	1	4			2	1		3	9	4	1	1	7	3		4	6	1	1	2		9	2	7	4	9
Replacement																										

BullSO, University of South Florida - Repaired practice edition

Division C | Codebusters | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

Name: _____ Date: _____

repair-case. (2 points) Authored written substitute for unavailable source material: Which operation preserves the exact letter counts while changing order?

A. Monoalphabetic letter replacement

B. Deleting spaces and letters

C. A numerical hash

D. Transposition

Answer: _____